HACKURITY 2026 · PS-6

Chain of Custody

Logs that notice tampering as it starts, prove what changed, and lock the scene.

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01 · THE PROBLEM

Once an attacker has root, the logs stop telling the truth.

Logs are the only witness

After a breach, logs are how a SOC or auditor reconstructs what happened. Banks, fintech and incident response all depend on them.

Root can rewrite anything

Delete lines, edit amounts, insert fake entries, truncate files. With root, the attacker can also hide the fact they did it.

Detection comes too late

Most integrity checks run after the fact. By then the evidence is gone and the attacker has already left.

02 · OUR ANGLE

Don't just detect tampering. Catch it, prove it, freeze it.

Catch

Before any log changes, flag the setup: agent stopped, file swapped, interface down. Layer 1.

Prove

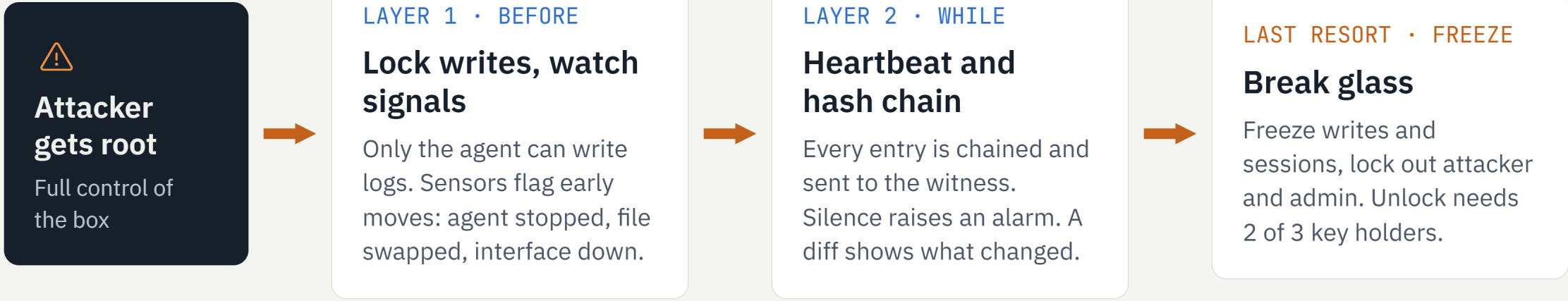
While it happens, show which entries changed, how, and how many are unverified after an outage. Layer 2.

Freeze

As a last resort, lock the box out and keep the evidence on the witness. Layer 3.

One agent, any platform, with per-platform profiles · SOC view + Auditor view

Three layers: before, while, and last resort



03 · LAYERS IN DETAIL

What each layer does, step by step

LAYER 1 · BEFORE

Sense and deter

- Only the agent's user can write log files (AppArmor or SELinux)
- Sensors watch the agent process, log inode and size, and interface state
- Command gate blocks common log tools as a visible speed bump

Limit: root can bypass this. Layer 2 covers that.

LAYER 2 · WHILE

Verify and prove

- Each entry is SHA-256 hashed and chained to the one before
- Entries are numbered and shipped to the witness as they're written
- A heartbeat every few seconds carries the latest number and hash
- Witness silence raises an alarm and records the last number received
- A diff finds the first entry that no longer matches

Limit: entries written during an outage stay unverified.

LAST RESORT · FREEZE

Freeze and preserve

- Triggered when Layer 2 confirms tampering on a host the attacker controls
- Freezes writes and sessions on the box
- Locks out the attacker and the admin account
- Evidence stays on the witness, which has no SSH
- Unlock needs 2 of 3 Shamir key holders

Limit: a false positive still needs a manual unlock.

How we build it

Component	Stack and technique
Agent + runtime	Python agents in Docker containers; one codebase with per-platform profiles
Layer 1 · Before	AppArmor or SELinux policy on log files; sensors for agent process, log inode and size, interface state
Layer 2 · While	SHA-256 hash chain, HMAC with forward-secure keys, sequence numbers, short heartbeats, witness silence alarm, per-entry diff
Layer 3 · Last resort	Freeze writes and sessions; evidence kept on the witness with no SSH; Shamir 2-of-3 split unlock key
Interfaces	SOC view for live alerts; Auditor view for verification reports

05 · SCALING

Where it goes next

More platforms

New profiles for banking cores, SaaS ops, critical infrastructure and blockchain audit firms.

Distributed witnesses

Several witnesses in a quorum, so no single node going down or getting cut off loses evidence.

SIEM integration

Feed tamper alerts straight into the tools SOC teams already use.

Compliance reports

One-click verified audit trails for regulators and external auditors.

Thank you · Questions?